Narendra Putra Bhaskoro
2510514022

No
Date

Soal 2 : Enkripsi Citra Logo Toko dengan Invers Matriks 3x3

Parameter Personalisasi
$N = 22$
$a = 2$
$b = 2$
$k = 5$
$\theta_0 = 70°$
$a = 0,003$

Matriks kunci A (3x3)

$$A = \begin{vmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{vmatrix} \Rightarrow A = \begin{vmatrix} 3 & 2 & 2 \\ 1 & 4 & 2 \\ 2 & 1 & 5 \end{vmatrix}$$

(a.) det (A) menggunakan ekspansi kofaktor baris pertama

$- A_{11} M_{11} = +3 . \begin{vmatrix} 4 & 2 \\ 1 & 5 \end{vmatrix} = \cancel{30}\ 3 . 18 = 54$

$- A_{12} M_{12} = -2 . \begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix} = -2 . 1 = -2$

$- A_{13} . M_{13} = +2 . \begin{vmatrix} 1 & 4 \\ 2 & 1 \end{vmatrix} = +2 . -7 = -14$

$\text{Det}(A) = 54 + (-2) + (-14)$
$= 38$

(b.) Matriks minor M_{ij} => Det (M_{ij})

$M_{11} = \begin{vmatrix} 4 & 2 \\ 1 & 5 \end{vmatrix} \Rightarrow 20 - 2 = 18$

$M_{12} = \begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix} \Rightarrow 5 - 4 = 1$

$M_{13} = \begin{vmatrix} 1 & 4 \\ 2 & 1 \end{vmatrix} \Rightarrow 1 - 8 = -7$

$M_{21} = \begin{vmatrix} 2 & 2 \\ 1 & 5 \end{vmatrix} \Rightarrow 10 - 2 = 8$

$M_{22} = \begin{vmatrix} 3 & 2 \\ 2 & 5 \end{vmatrix} \Rightarrow 15 - 4 = 11$

$M_{23} = \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} \Rightarrow 3 - 4 = -1$

$M_{31} = \begin{vmatrix} 2 & 2 \\ 4 & 2 \end{vmatrix} \Rightarrow 4 - 8 = -4$

$M_{32} = \begin{vmatrix} 3 & 2 \\ 1 & 2 \end{vmatrix} \Rightarrow 6 - 2 = 4$

$M_{33} = \begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix} \Rightarrow 12 - 2 = 10$

JOYKO 36 Lines, 6 mm

Scanned with CamScanner

$$\Rightarrow M = \begin{vmatrix} 18 & 1 & -7 \\ 8 & 11 & -1 \\ -4 & 4 & 10 \end{vmatrix}$$

//

JOYKO® 36 Lines, 6 mm

Scanned with CamScanner

No
Date

C. $\Rightarrow \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$

masukkan matriks minor ke dalam situ

$$C \Rightarrow \begin{bmatrix} 18 & 1 & -7 \\ 8 & 11 & -1 \\ -4 & 4 & 10 \end{bmatrix} \Rightarrow \begin{bmatrix} 18 & -1 & -7 \\ -8 & 11 & 1 \\ -4 & -4 & 10 \end{bmatrix}$$

kemudian, buat matriks Adjoin (Adj (A))

$$(Adj(A)) = C^T$$

$$C = \begin{bmatrix} 18 & -1 & -7 \\ -8 & 11 & 1 \\ -4 & -4 & 10 \end{bmatrix} \Rightarrow C^T = \begin{bmatrix} 18 & -8 & -4 \\ -1 & 11 & -4 \\ -7 & 1 & 10 \end{bmatrix}$$

Matriks Invers (A^{-1})

$$(A^{-1}) = \frac{1}{\det(A)} \cdot Adj(A)$$

$$= \frac{1}{38} \cdot \begin{bmatrix} 18 & -8 & -4 \\ -1 & 11 & -4 \\ -7 & 1 & 10 \end{bmatrix}$$

$$= \begin{bmatrix} 18/38 & -8/38 & -4/38 \\ -1/38 & 11/38 & -4/38 \\ -7/38 & 1/38 & 10/38 \end{bmatrix} \Rightarrow \begin{bmatrix} 9/19 & -4/19 & -2/19 \\ -1/38 & 11/38 & -2/19 \\ -7/38 & 1/38 & 5/19 \end{bmatrix}$$

//

Scanned with CamScanner

No
Date

(e.) Studi Kasus Citra

$$P = \begin{bmatrix} 120 & 50+N & 200 \\ 80 & a+2 & 10+N \\ 30+N & 90 & 170 \end{bmatrix} \Rightarrow \begin{bmatrix} 120 & 72 & 200 \\ 80 & 4 & 32 \\ 52 & 90 & 170 \end{bmatrix}$$

- Enkripsi Manual ($E = A \cdot P$)

$$= \begin{bmatrix} 3 & 2 & 2 \\ 1 & 4 & 2 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} 120 & 72 & 200 \\ 80 & 4 & 32 \\ 52 & 90 & 170 \end{bmatrix} \Rightarrow$$

Baris 1 : $360 + 160 + 104 = 624$
$216 + 8 + 180 = 404$
$600 + 64 + 340 = 1004$

Baris 2 : $120 + 320 + 104 = 544$
$72 + 16 + 180 = 268$
$200 + 128 + 340 = 668$

Baris 3 : $240 + 80 + 260 = 580$
$144 + 8 + 450 = 598$
$400 + 32 + 850 = 1282$

$$E = \begin{bmatrix} 624 & 404 & 1004 \\ 544 & 268 & 668 \\ 580 & 598 & 1282 \end{bmatrix}$$

- Deskripsi Manual ($P' = A^{-1} \cdot E$)

$$P' = \frac{1}{38} \begin{bmatrix} 18 & -8 & -4 \\ -1 & 11 & -4 \\ -7 & 1 & 10 \end{bmatrix} \begin{bmatrix} 624 & 404 & 1004 \\ 544 & 268 & 668 \\ 580 & 598 & 1282 \end{bmatrix}$$

Coba ambil satu ~~angka~~ baris pertama

$\Rightarrow (18 \times 624) + (-8 \times 544) + (-4 \times 580)$

$= 11232 - 4352 - 2320$

$= 4560 \Rightarrow \frac{4560}{38} = 120$ (kembali ke nilai awal)

$$P' = \begin{bmatrix} 4560/38 & 2736/38 & 7600/38 \\ 2040/38 & 152/38 & 1216/38 \\ 1976/38 & 3420/38 & 6460/38 \end{bmatrix} = \begin{bmatrix} 120 & 72 & 200 \\ 80 & 4 & 32 \\ 52 & 90 & 170 \end{bmatrix} \Rightarrow \text{terbukti } P' = P$$

- Apa yang terjadi jika det (A) = 0

Jika determinan = 0, maka matriks itu merupakan matriks singular. Dalam enkripsi citra, hal tersebut dapat menyebabkan proses enkripsi bersifat irreversibel / tidak bisa dibalik, Saat sistem mencoba proses dekripsi ($P' = A^{-1} \cdot E$), akan terjadi error matematis akibat pembagian dengan angka nol. Akibatnya, citra logo PT. Bela Negara tidak akan bisa direstorasi sama sekali dan datanya akan rusak atau hilang secara permanen.

JOYKO® 36 Lines, 6 mm

Scanned with CamScanner